

EDUCATION

Ph.D in Computer Science and Information Engineering (CSIE) , <i>National Taiwan University (NTU), Taiwan</i> Research Domain: Storage system, computer architecture, intermittent computing, and robotics - Supervised by Prof. Tei-Wei Kuo and Prof. Yuan-Hao Chang	02/2020 – Present
M.S. and B.S. in CSIE , <i>National Cheng Kung University (NCKU), Taiwan</i> Research Domain: Intermittent Computing - Supervised by Prof. Chia-Heng Tu and by Prof. Yuan-Hao Chang	09/2014 – 01/2020

TECHNICAL EXPERIENCE

Research Assistant / Intermittent Computing Systems (Academia Sinica) - Supervisor: Prof. Chia-Heng Tu and Prof. Yuan-Hao Chang - Research Domain: Intermittent computing, hyperdimensional computing and deep neural network	01/2020 – 12/2025
Visiting Researcher / In-Storage Computing (CityU, HK) - Supervisor: Prof. Tei-Wei Kuo and Prof. Chun Jason Xue - Research Domain: In-storage computing	04/2022 – 07/2022
Teaching Assistant / Compiler and Virtual Machine (NCKU) Lex and Yacc, Linux container, Intel Software Guard Extensions (SGX) with SCONE	02/2019 – 07/2019
Teaching Assistant / Real-Time System (NTU) Linux kernel real-time scheduler implementation	02/2024 – 06/2025
Undergraduated Project (NCKU) - Location-based intelligent catering refill system on Raspberry Pi and Beacons (with RadioFlux) - Memory allocation project with Supermalloc, PTMalloc2 and Daniel Micay's Allocator (Supervised by Jserv) - Raytracing optimization project with Perf, gprof, PThread, SIMD, AutoFDO, and OpenTuner (Supervised by Jserv)	02/2017 – 01/2018

PUBLICATION (JOURNAL PAPERS)

- Ssu-Hao Tsai, Wen Sheng Lim, Liang-Chi Chen, Tei-Wei Kuo, Yuan-Hao Chang, "GraphISC: Enabling Out-of-Core Graph Processing with Distributed Computational Storage," *accepted and to appear in IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD) (Integrated with ACM/IEEE CASES'26)*
- Wen Sheng Lim, Shu-Ting Cheng, Ya-Tung Tsai, Chia-Heng Tu, and Yuan-Hao Chang, "Timing-Constrained Composable Inference for Intermittent Systems using Reinforcement Learning," *accepted and to appear in ACM Transactions on Embedded Computing Systems (TECS)*.
- Wen Sheng Lim, Yu-Cheng Chen, Yu-Hsuan Chu, Chia-Heng Tu, Yuan-Hao Chang, "iSAFE: Enabling Evenness of Data Freshness in Multipriority Networked Intermittent Systems," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)*, vol. 44, no. 6, pp. 2093–2104, Jun. 2025.
- Jalil Boukhobza, Pierre Olivier, Wen Sheng Lim, Liang-Chi Chen, Yun-Shan Hsieh, Shin-Ting Wu, Chien-Chung Ho, Po-Chun Huang, Yuan-Hao Chang, "A Survey on Flash-Memory Storage Systems: A Host-Side Perspective," *ACM Transactions on Storage (TOS)*, vol. 21, no. 3, pp. 24:1–24:59, Jun. 2025.
- Wen Sheng Lim, Chia-Heng Tu, Chun-Feng Wu, and Yuan-Hao Chang, "iCheck: Progressive Checkpointing for Intermittent Systems," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)*, vol. 40, no. 11, pp. 2224–2236, Nov. 2021.

PUBLICATION (CONFERENCE PAPERS)

- Ya-Tung Tsai, Wen Sheng Lim, Yu-Cheng Chen, Chia-Heng Tu and Yuan-Hao Chang, "Progress Gambit: Rethinking Checkpointing for Intermittent Hyperdimensional Computing on Ultra-Low-Power Devices," *ACM/IEEE International Conference on Computer-Aided Design (ICCAD)*, San Jose, California, USA, Nov. 8–12, 2026. **(Top Conference)**
- Ssu-Hao Tsai, Wen Sheng Lim, Liang-Chi Chen, Tei-Wei Kuo, Yuan-Hao Chang, "GraphISC: Enabling Out-of-Core Graph Processing with Distributed Computational Storage," *ACM/IEEE International Conference on Compilers, Architecture, and Synthesis for Embedded Systems (CASES)*, Barcelona, Spain, Oct. 4–9, 2026. **(Top Conference)**

3. Yi-Sia Gao, Wen Sheng Lim, Tei-Wei Kuo and Yuan-Hao Chang, “ReMemTier: Rethinking Memory Tiering for CXL-based SSD via Large Granularity Memory Access,” *ACM/IEEE The Chips to Systems Conference (DAC)*, Long Beach, California, USA, Jul. 26–29, 2026. **(Top Conference)**
4. Wen-Hsin Liu, Chieh-Lin Tsai, Wen Sheng Lim, Ray-Ting Huang, Han-Wen Hu, Tei-Wei Kuo and Yuan-Hao Chang, “FlashHD: A Flash-Based In-Storage Hyperdimensional Computing Framework for Hierarchical Sequence Matching,” *ACM/IEEE The Chips to Systems Conference (DAC)*, Long Beach, California, USA, Jul. 26–29, 2026. **(Top Conference)**
5. Jhao-Ying Chen, Wen Sheng Lim, Tei-Wei Kuo, and Yuan-Hao Chang, “STAR: High-DoF Robotic Manipulation for Memory-Constrained NN Accelerator,” *ACM/IEEE Design, Automation and Test in Europe Conference (DATE)*, Verona, Italy, Apr. 20–22, 2026. **(Top Conference)**
6. Wen Sheng Lim, Chia-Heng Tu, and Yuan-Hao Chang, “A Volunteer Computing Framework for IoT Devices using Intermittent Computing,” *ACM Symposium on Applied Computing (SAC)*, Thessaloniki, Greece, Mar. 23–27, 2026.
7. Yu-Hong Lai, Chieh-Lin Tsai, Wen Sheng Lim, Han-Wen Hu, Tei-Wei Kuo, and Yuan-Hao Chang, “ReCross: Efficient Embedding Reduction Scheme for In-Memory Computing using ReRAM-Based Crossbar,” *ACM Symposium on Applied Computing (SAC)*, Thessaloniki, Greece, Mar. 23–27, 2026.
8. Meng-Chia Lee, Wen Sheng Lim, Yuan-Hao Chang, and Tei-Wei Kuo, “SARA: A Stall-Aware Memory Allocation Strategy for Mixed-Criticality Systems,” *ACM/IEEE Asia and South Pacific Design Automation Conference (ASP-DAC)*, Hong Kong, Jan. 19–22, 2026.
9. Shu-Ting Cheng, Wen Sheng Lim, Chia-Heng Tu and Yuan-Hao Chang, “TRAIN: A Reinforcement Learning Based Timing-Aware Neural Inference on Intermittent Systems,” *ACM/IEEE International Conference on Computer-Aided Design (ICCAD)*, San Francisco, California, USA, Oct. 29 – Nov. 2, 2023. **(Top Conference)**
10. Hao-Jan Huang, Wen Sheng Lim, Chia-Heng Tu, Chun-Feng Wu, and Yuan-Hao Chang, “Data Freshness Optimization on Networked Intermittent Systems,” *ACM/IEEE Design, Automation and Test in Europe Conference (DATE)*, Antwerp, Belgium, Apr. 17–19, 2023. **(Top Conference)**